

ADIOLOGY

Search Ads Intelligence Platform

System Design Document

Comprehensive Technical Architecture & Specification

Document Version: 2.0
Generated: February 14, 2026
Classification: Internal / Confidential

<https://adiology.io>

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1. Executive Summary

Adiology is a comprehensive Google Ads campaign intelligence platform designed to automate and streamline the creation, management, and optimization of search advertising campaigns. The platform provides end-to-end campaign management from keyword research through ad generation, CSV export in Google Ads Editor format, and direct push to Google Ads via OAuth integration.

The platform serves digital marketing professionals, agencies, and businesses seeking to maximize their search advertising efficiency through AI-powered automation, competitive intelligence, and click fraud protection.

Core Value Propositions

- Automated campaign structure generation (SKAG, STAG, Intent-Based, Alpha-Beta)
- AI-powered keyword generation and ad copy creation
- Direct Google Ads integration via OAuth for campaign push/update
- Click fraud protection with IP blocking and Google Ads exclusion sync
- Competitive intelligence through ad transparency research
- Domain monitoring with SSL, DNS, and WHOIS tracking
- Anonymous proxy email for competitive research

2. System Architecture Overview

Adiology follows a monolithic full-stack architecture with clear separation of concerns between the frontend React application and the backend Hono API server. Both are deployed together but serve distinct responsibilities.

High-Level Architecture

The system is structured as follows:

- Client Layer: React SPA served via Vite dev server (port 5000)
- API Layer: Hono (Node.js/TypeScript) backend server (port 3001)
- Data Layer: PostgreSQL (Neon-backed) with Drizzle ORM
- Auth Layer: Clerk authentication with JWT tokens
- Payment Layer: Stripe subscriptions with webhook handlers
- AI Layer: OpenAI GPT integration for content generation
- Email Layer: Resend API for transactional and marketing emails

Request Flow

1. User interacts with the React frontend
2. Frontend sends API requests with Clerk JWT tokens
3. Vite dev server proxies `/api/*` requests to Hono backend on port 3001
4. Hono middleware validates JWT, extracts `userId`
5. Route handler processes business logic with Drizzle ORM
6. Response returns through the proxy chain to the client

3. Technology Stack

Frontend

Framework	React 18 with TypeScript
Build Tool	Vite 6.x
UI Library	Radix UI + Tailwind CSS
State Management	React Context + Hooks
Routing	Wouter (lightweight router)
Animation	Framer Motion
Charts	Recharts
Icons	Lucide React
HTTP Client	Native fetch API
Form Validation	Zod schemas

Backend

Runtime	Node.js with TypeScript (tsx)
Web Framework	Hono
Database ORM	Drizzle ORM
Validation	Zod + drizzle-zod
Auth	Clerk JWT verification
API Format	RESTful JSON APIs

Database & Infrastructure

Database	PostgreSQL (Neon-backed via Replit)
Hosting	Replit (development) / Custom domain
CDN/Proxy	Vite dev server proxy
SSL	Managed via Replit/Cloudflare

Third-Party Services

Authentication	Clerk
Payments	Stripe
AI/NLP	OpenAI GPT-4
Email	Resend
Analytics	Google Analytics (gtag.js)
Ads Platform	Google Ads API v18
Temp Email	temp-mail.io API

4. Frontend Architecture

Directory Structure

```
src/
  App.tsx                # Main app with routing
  main.tsx               # Entry point
  components/            # Feature components (~95 files)
    ui/                  # Shared UI primitives (Radix)
    admin/               # Admin panel components
    ai-chat/             # AI assistant chat
    feature-pages/       # Marketing feature pages
    onboarding/          # User onboarding flow
  contexts/              # React contexts (Auth, Theme)
  hooks/                 # Custom React hooks
  utils/                 # Utility functions & API helpers
  modules/               # Feature modules (community)
  schemas/               # Zod validation schemas
  styles/                # Global CSS styles
```

Key Frontend Components

Campaign Builder (CampaignBuilder3.tsx)

A 7-step wizard for building Google Ads campaigns: URL Analysis -> Structure Selection -> Keyword Generation -> Ad Generation -> Geo Targeting -> Preview -> CSV Export/Push.

Click Guard (ClickGuard.tsx)

Click fraud protection dashboard with tabs for: Domain Management, Live Traffic Monitor, Analytics Dashboard, Protection Rules, and Blocked IPs with Google Ads IP exclusion sync.

Google Ads Push (GoogleAdsPushButton.tsx)

OAuth-based Google Ads integration for pushing/updating campaigns directly. Supports auto-detection of accounts with manual Customer ID fallback.

Domain Monitoring (DomainMonitoring.tsx)

Tracks domain expiry, SSL certificates, and DNS records with email alert configuration.

Proxy Mail (TempMail.tsx)

Anonymous disposable email generation for competitive intelligence research.

Keyword Tools

Suite of keyword tools: Keyword Planner, Long Tail Generator, Negative Keywords Builder, Keyword Mixer, and Saved Lists management.

Super Admin Panel (SuperAdminPanel.tsx)

Comprehensive admin console for user management, subscription management, system logs, email marketing automation, database administration, and security configuration.

5. Backend Architecture

Server Configuration

The backend runs as a Hono application on Node.js, using tsx for TypeScript execution. It serves API routes and includes CORS configuration, error handling, and request logging.

Route Modules

Account (/api/account)

User profile management, settings, preferences

Admin (/api/admin)

Admin-only operations, user management, system config

Super Admin (/api/superadmin)

Super admin operations, database CRUD, email automation

Click Guard (/api/clickguard)

Domain tracking, visitor logging, IP blocking, fraud detection

Google Ads (/api/google-ads)

OAuth flow, campaign push/update, IP exclusion sync

Stripe (/api/stripe)

Subscription management, webhook handlers, payment processing

Community (/api/community)

Discourse SSO integration, forum topics and categories

Domains (/api/domains)

Domain monitoring, WHOIS/SSL/DNS checking, alerts

Tasks (/api/tasks)

Task management CRUD, project organization

Organizations (/api/organizations)

Multi-tenant organization management

Invites (/api/invites)

Organization invite system with codes

Seats (/api/organization)

Seat-based billing and member management

Temp Mail (/api/tempmail)

Disposable email proxy via temp-mail.io

User (/api/user)

User sync, profile management

Promo (/api/promo)

Promotional trial management

Tickets (/api/tickets)

Support ticket CRUD with admin replies (not yet registered)

Middleware Stack

- CORS: Allow all origins with credential support
- Logger: Request/response logging via Hono logger
- Auth: JWT verification via Clerk tokens (per-route)
- Admin Auth: Separate admin token middleware for admin routes
- Error Handler: Global error boundary returning JSON errors

6. Database Schema

The database uses PostgreSQL (Neon-backed) managed through Drizzle ORM. All tables use UUID primary keys (except tasks/task_projects which use serial) and include indexing for performance.

Core Tables

users

User accounts and profiles

id (text PK), email, fullName, avatarUrl, role, subscriptionPlan
subscriptionStatus, stripeCustomerId, stripeSubscriptionId, aiUsage
isBlocked, lastSignIn, passwordHash, emailVerified, cardValidated, selectedPlan

subscriptions

Stripe subscription tracking

id (uuid PK), userId, stripeCustomerId, stripeSubscriptionId, stripePriceId
planName, status, currentPeriodStart/End, cancelAtPeriodEnd, trialStart/End

payments

Payment history and receipts

id (uuid PK), userId, subscriptionId, stripePaymentIntentId, stripeInvoiceId
amountCents, currency, status, paymentMethodType, description, receiptUrl

campaign_history

Saved campaign builds

id (uuid PK), userId, workspaceId, type, name, data (JSONB), status
googleAdsId, googleAdsPushStatus, googleAdsPushedAt

workspaces

Workspace/multi-tenant containers

id (uuid PK), name, slug, ownerId, isAdminWorkspace, settings (JSONB)

workspace_members

Workspace membership

id (uuid PK), workspaceId, userId, role, status

Click Guard Tables

click_guard_domains

Tracked domains for click protection

id (uuid PK), userId, domain, siteId (unique), verified, settings (JSONB)

click_guard_visitors

Visitor tracking logs

id (uuid PK), siteId, ipAddress, userAgent, fingerprint, country, city
deviceType, browser, os, isProxy, isVpn, isBot, isTor, botScore, threatLevel
clickCount, mouseMovements, timeOnPage, blocked

click_guard_blocked_ips

Blocked IP addresses

id (uuid PK), siteId, ipAddress, reason, autoBlocked, expiresAt

click_guard_fraud_events

Detected fraud events

id (uuid PK), siteId, visitorId, eventType, severity, ipAddress, details (JSONB)

click_guard_ip_push_log

Google Ads IP exclusion push history

id (uuid PK), siteId, userId, googleAdsCustomerId, campaignIds (JSONB)
ipsCount, ipsPushed (JSONB), status, errorMessage, pushedAt

Domain Monitoring Tables

monitored_domains

Tracked domains for monitoring

id (uuid PK), userId, domain, registrar, expiryDate, nameServers (JSONB)
sslIssuer, sslExpiryDate, sslData (JSONB), dnsRecords (JSONB)
alertDays (JSONB), alertsEnabled, alertEmail, status

domain_snapshots

Point-in-time domain snapshots

id (uuid PK), domainId (FK), snapshotType, data (JSONB), changes (JSONB)

domain_alerts

Domain alert history

id (uuid PK), domainId (FK), alertType, message, daysUntilExpiry, sentAt, acknowledged

Other Tables

emails - Transactional email tracking with delivery/open/click status

email_logs - Email sequence and marketing email logs

email_sequence_progress - Email automation sequence tracking per user

audit_logs - System audit trail with action/resource/IP tracking

security_rules - IP blocking, rate limiting rules

templates - Website/landing page templates

saved_sites - User-saved website builds

published_websites - Vercel-deployed website records

feedback - User feedback and ratings

forms / form_submissions - Custom form builder and submission storage

task_projects / tasks - Task management with projects and priorities

organizations / org_members / org_invites - Multi-tenant organization system

workspace_projects / project_items - Workspace project organization

conversations / messages - AI chat conversation history

user_notifications - In-app notification system

ad_search_requests - Google Ads transparency search history

promo_trials - Promotional trial tracking

support_tickets - Customer support ticket system

google_ads_tokens - Google Ads OAuth token storage

kv_store - General key-value storage

invoices - Invoice tracking with Stripe integration

documentation_images - Help center documentation images

7. API Endpoints

Authentication & Account

GET /api/account/me - Get current user profile
POST /api/user/sync - Sync user from Clerk to database
GET /api/google-ads/auth/url - Get Google Ads OAuth URL
GET /api/google-ads/auth/callback - OAuth callback handler
GET /api/google-ads/auth/status - Check Google Ads connection status
GET /api/google-ads/auth/accounts - List accessible Google Ads accounts
POST /api/google-ads/auth/disconnect - Disconnect Google Ads

Campaign Management

GET /api/campaign-history - List user campaign history
POST /api/campaign-history - Save campaign to history
POST /api/google-ads/push - Push campaign to Google Ads
POST /api/google-ads/update - Update existing Google Ads campaign
GET /api/google-ads/campaigns - List Google Ads campaigns
POST /api/analyze-url - Analyze URL for campaign building

Click Guard

POST /api/clickguard/domains - Register domain for tracking
GET /api/clickguard/domains/:userId - List tracked domains
GET /api/clickguard/stats/:siteId - Get site analytics stats
GET /api/clickguard/visitors/:siteId - List visitor logs
POST /api/clickguard/track - Record visitor event
GET /api/clickguard/blocked-ips/:siteId - List blocked IPs
POST /api/clickguard/block-ip - Block an IP address
DELETE /api/clickguard/unblock-ip/:id - Unblock an IP
GET /api/clickguard/export-blocked-ips/:siteId - Export IPs as CSV
POST /api/clickguard/push-ip-exclusions - Push IPs to Google Ads

Stripe & Billing

POST /api/stripe/create-checkout - Create Stripe checkout session
POST /api/stripe/webhook - Stripe webhook handler
GET /api/stripe/subscription/:userId - Get subscription details
POST /api/stripe/cancel - Cancel subscription
POST /api/stripe/create-portal-session - Stripe billing portal

Domain Monitoring

GET /api/domains/:userId - List monitored domains
POST /api/domains - Add domain to monitoring
PUT /api/domains/:id - Update domain settings
DELETE /api/domains/:id - Remove domain from monitoring
POST /api/domains/:id/check - Trigger domain check

Other Endpoints

GET/POST /api/tempmail/* - Proxy email operations
GET/POST /api/tasks/* - Task management CRUD
GET/POST /api/organizations/* - Organization management
GET/POST /api/invites/* - Invite system
GET/POST /api/community/* - Discourse SSO & forum proxy
GET/POST /api/admin/* - Admin operations
GET/POST /api/superadmin/* - Super admin operations
GET/POST /api/promo/* - Promo trial management
GET /api/health - Health check
GET /api/notifications/:userId - User notifications

8. Authentication & Authorization

Authentication Flow

Adiology uses Clerk as the primary authentication provider. The flow works as follows:

- User signs up/logs in via Clerk (email/password or social login)
- Clerk issues a JWT token stored in the browser session
- Frontend includes the JWT as Bearer token in API requests
- Backend verifies JWT using Clerk's public key
- User ID is extracted from the verified token for authorization

For Google Ads integration, a separate OAuth 2.0 flow connects users' Google Ads accounts. OAuth tokens (access + refresh) are stored in the google_ads_tokens table and automatically refreshed when expired.

Authorization Levels

Role: user	Default role. Access to all user-facing features based on subscription plan.
Role: admin	Access to admin panel, user management, and system configuration.
Role: superadmin	Full access including database administration, email automation, and security rules.

Subscription Tiers

Free	Limited campaign builds, basic features
Starter	Extended limits, keyword tools, domain monitoring
Professional	Full access, Click Guard, Google Ads push, priority support
Agency	Multi-user, organization management, seats, white-label options

Security Measures

- JWT token verification on all authenticated endpoints
- CORS configured with credential support
- Admin routes protected by separate admin auth middleware
- IP-based security rules for blocking suspicious activity
- Rate limiting via security rules table
- Content Security Policy headers
- Stripe webhook signature verification

9. Feature Modules

9.1 Campaign Builder

The flagship feature - a 7-step wizard that guides users through creating a complete Google Ads campaign:

- Step 1: URL Input & AI Analysis - Analyzes target website using Cheerio scraper
- Step 2: Campaign Structure - Choose SKAG, STAG, Intent-Based, or Alpha-Beta
- Step 3: Keyword Generation - AI-powered keyword suggestions with match types
- Step 4: Ad Generation - RSA ads with headlines/descriptions following Google policies
- Step 5: Geo Targeting - Country, state, city, and ZIP code targeting
- Step 6: Preview - Campaign review with ad preview
- Step 7: Export - CSV download (Google Ads Editor format) or direct Push to Google Ads

9.2 Click Guard

Click fraud protection system providing:

- Lightweight JavaScript tracking script for websites
- Bot detection engine with fingerprinting
- Live traffic monitor showing real-time visitors
- Analytics dashboard with charts and threat analysis
- Automatic and manual IP blocking
- Google Ads IP exclusion sync - push blocked IPs as campaign exclusions
- CSV export of blocked IP lists

9.3 Domain Monitoring

Comprehensive domain health tracking:

- WHOIS data monitoring with registrar and expiry tracking
- SSL certificate monitoring with validity and issuer info
- DNS record tracking (A, AAAA, MX, TXT, CNAME, NS)
- Configurable email alerts at 30, 15, 7, and 1 day before expiry
- Point-in-time snapshots for change tracking

9.4 Proxy Mail

Anonymous email system for competitive research:

- Generate disposable email addresses instantly
- Subscribe to competitor newsletters anonymously
- Live inbox with message viewing
- Auto-expiry for privacy protection
- Email history tracking

9.5 Keyword Intelligence

Suite of keyword research tools:

- Keyword Planner - AI-powered keyword suggestions
- Long Tail Keywords - Extended keyword variations
- Negative Keywords Builder - Identify irrelevant keywords
- Keyword Mixer - Combine keyword components
- Saved Lists - Organize and manage keyword collections

9.6 Task Manager

Built-in project and task management:

- Project-based organization with color coding
- Task priorities (low, medium, high)
- Due dates and completion tracking
- Today view for focused task management

9.7 AI Blog Generator

AI-powered content creation tool for generating long-form blog posts with configurable parameters for topic, length, tone, and SEO optimization.

9.8 Campaign Presets

Pre-built campaign templates for common industries and use cases, allowing users to start with proven structures and customize from there.

10. Third-Party Integrations

Clerk Authentication

Handles all user authentication including email/password, social login (Google, GitHub), email verification, and session management. Integration via Clerk publishable key on frontend and JWT verification on backend.

Stripe Payments

Complete payment infrastructure:

- Three subscription tiers (Starter, Professional, Agency)
- 7-day free trial support
- Stripe Checkout for secure payment processing
- Webhook handlers for subscription lifecycle events
- Customer portal for self-service billing management
- Invoice generation and receipt tracking

Google Ads API v18

Direct Google Ads integration:

- OAuth 2.0 authorization flow with token management
- Auto-detection of accessible customer accounts
- Manual Customer ID fallback for test mode compatibility
- Campaign creation with ad groups, keywords, and ads
- Campaign update support for existing campaigns
- Geo-targeting with location criteria
- IP exclusion management for click fraud protection
- Campaign listing via searchStream API

OpenAI

AI integration for intelligent content generation:

- URL analysis for marketing insights
- Keyword generation and expansion
- Ad copy writing following Google Ads policies
- Blog content generation
- AI chat assistant for campaign optimization advice

Resend

Email delivery service for:

- Transactional emails (welcome, verification, receipts)
- Email marketing sequences with automation
- Open/click/bounce tracking
- Template-based email rendering

Google Analytics

Client-side analytics via gtag.js for tracking user behavior, page views, and conversion events.

Discourse Community

Forum integration via SSO for community discussions, knowledge sharing, and support.

11. Security Architecture

Application Security

- JWT-based authentication with Clerk token verification
- Role-based access control (user, admin, superadmin)
- API key authentication for admin endpoints
- CORS configuration with credential support
- Content Security Policy headers
- Input validation via Zod schemas
- SQL injection prevention via Drizzle ORM parameterized queries

Payment Security

- PCI-DSS compliance via Stripe Checkout (no card data on server)
- Stripe webhook signature verification
- Server-side subscription status verification

Data Protection

- Google Ads OAuth tokens encrypted at rest in database
- Environment variables for all secrets and API keys
- No secrets exposed in client-side code
- Audit logging for administrative actions
- IP-based security rules for threat mitigation

Click Fraud Protection

- Bot detection via behavioral analysis (mouse movements, timing)
- Device fingerprinting for visitor identification
- VPN/Proxy/Tor detection flags
- Automated IP blocking based on threat score
- Google Ads IP exclusion sync for active protection

12. Deployment & Infrastructure

Development Environment

- Replit-based development with NixOS environment
- Frontend: Vite dev server on port 5000 with HMR
- Backend: Hono server on port 3001 via tsx
- Database: Replit PostgreSQL (Neon-backed)
- Hot reload for both frontend and backend changes

Production Configuration

- Frontend: Vite production build to /build directory
- Backend: Hono serving static files + API routes
- Custom domain: adiology.io
- SSL: Managed via hosting platform

- Database: Production PostgreSQL with connection pooling

Environment Variables

- DATABASE_URL - PostgreSQL connection string
- CLERK_PUBLISHABLE_KEY / CLERK_SECRET_KEY - Authentication
- STRIPE_SECRET_KEY / STRIPE_WEBHOOK_SECRET - Payments
- GOOGLE_ADS_CLIENT_ID / CLIENT_SECRET / DEVELOPER_TOKEN - Google Ads
- OPENAI_API_KEY - AI features
- RESEND_API_KEY - Email delivery
- TEMP_MAIL_API_KEY - Proxy email service
- GOOGLE_ANALYTICS_ID - Analytics tracking